LX-6 User Manual

Six-voice digital drum synthesizer · AU / VST3 / standalone · version 0.1.0



LX-6 runs the sound engine of the open-source Erica Synths LXR drum machine firmware inside your DAW: the same oscillators, FM, noise, transients, envelopes, filters and effects, driven by a 16-step sequencer, a sound randomizer and, if you own one, a real LXR-02 over MIDI.

LX-6 is an independent product by Ticco Ross. It is not made, approved or endorsed by Erica Synths.

1. Installing

macOS (Apple Silicon and Intel, macOS 11 or later)

- Open LX-6-0.1.0-macOS.pkg. You can choose Audio Unit, VST3 and the standalone app.
- The installer is not notarized yet. If macOS refuses to open it, right-click the .pkg and choose Open, or allow it in System Settings > Privacy & Security.
- Plugins go to /Library/Audio/Plug-Ins/Components (AU) and /Library/Audio/Plug-Ins/VST3. Rescan plugins in your DAW.

Windows (64-bit)

- Run LX-6-0.1.0-Windows.exe. The VST3 goes to C:\Program Files\Common Files\VST3, the app to C:\Program Files\Ticco Ross\LX-6.

LX-6 is an instrument: put it on an instrument / MIDI track. In Ableton Live it appears under Plug-Ins > Ticco Ross.

2. Quick start

- Press play in your DAW. Pattern A holds a starter beat; the sequencer follows the DAW tempo and position.
- In the standalone app (or with the DAW stopped) press PLAY in the sequencer bar and drag the TEMPO display up/down.
- Click a voice strip to edit that voice. Click the coloured LED next to its name to hear it.
- Click RANDOM for a new kit, UNDO to go back.
- Play the voices from a keyboard or MIDI clip: C1–F#1 (notes 36–42).

3. The panel

Top bar

Control	What it does
LX-6 / gumroad link	Opens the Ticco Ross store.
KIT	Shows the kit name. Click for the kit menu: Init Kit, your saved kits, Save kit as..., Load kit file..., Show kits folder.
SAVE	Saves the current sound as a .lx6kit file (default folder: Documents/LX-6/Kits).
RANDOM · AMT · UNDO	Randomizes the whole kit. AMT sets how far: low = variations of the current kit, full = brand new sounds. Locked sections are kept. UNDO steps back (up to 32 times).
TEMPO	Shows the host tempo (HOST SYNC) or the internal tempo (INTERNAL). Drag up/down to change the internal tempo.
MASTER	Output level (−60 to +6 dB). Double-click: 0 dB. The meter shows the output.
LXR	Hardware editor settings for an Erica Synths LXR-02 (lit when active). See chapter 7.
i	About LX-6.

Voice strips

Six strips, one per voice: DRUM 1, DRUM 2, DRUM 3, SNARE, CYMBAL, HI-HAT. Each has a trigger LED (click to audition), an envelope picture (attack, decay, slope), COARSE pitch, DECAY, PAN, a volume fader and M (mute) / S (solo). The selected strip is outlined; its voice is shown in the editor. Double-click any knob to reset it to the Init kit value.

Voice editor

Two pages for the selected voice. Each section has a dice (randomize just this section) and a lock (keep this section when you randomize the kit or the voice). The dice in the header randomizes the whole voice. Hover a knob to see its value.

Page / section	Parameters
SOUND · OSCILLATOR	Wave (sine, triangle, saw, square, noise, PWM), coarse, fine, PWM. Drums: modulation wave, FM amount, FM frequency, FM/Mix mode. Snare: noise frequency and noise mix. Cymbal and hi-hat: waves 2 and 3, two modulation oscillators (frequency and gain).
SOUND · PITCH MOD	Drums and snare: pitch envelope decay, slope and amount.
SOUND · TRANSIENT	Cymbal and hi-hat (in place of pitch mod): transient wave, level and frequency.
SOUND · FILTER	Type (LP, HP, BP, UBP, notch, peak, LP2, off), cutoff, resonance, filter drive.

SOUND · AMP ENVELOPE	Attack, decay (hi-hat: closed and open decay), slope. Snare and cymbal: repeat.
MOD · TRANSIENT	Drums and snare: transient wave, level and frequency.
MOD · VOICE	Volume, pan, drive, S.RATE (sample-rate reduction: 127 is clean, lower is crunchier), output (St1, St2, L1, R1, L2, R2 or FX bus). Locked by default so random kits keep their mix.
MOD · VELOCITY	Velocity amount, velocity destination (any parameter), velocity to volume on/off.

FX bus and LFO (right column)

The FX bus is fed by every voice whose OUTPUT is set to FX. Choose OFF, DRV (drive), RNG (ring modulator), CMP (compressor) or DEL (delay). DRY/WET and P1–P4 change meaning with the effect (delay: time, range, tone, feedback; compressor: attack, release, threshold, gain; drive: drive and tone shaping; ring: frequency). Extra menus appear for the chosen effect (distortion type, ratio, ring wave, delay or ping-pong). FX OUT sets where the effect goes. S.RATE ALL reduces the sample rate of everything.

The LFO block belongs to the selected voice: RATE, DEPTH, OFFSET, SHAPE (sine, triangle, saw up/down, square, random, exponential up/down), SYNC (off or 4/1 … 1/32 of the tempo), DEST (any parameter of any voice) and RETRIG (restart the LFO when a voice plays).

4. Sequencer

- A–H: eight patterns. Click to play and edit a pattern; while the sequencer runs it switches at the end of the bar (the button blinks until then). Shift-click: edit without switching. Alt-click: copy the pattern you are editing to that slot.
- D1 D2 D3 SN CY HH OH: the track you are editing (OH = open hi-hat, which chokes the closed hi-hat). Selecting a voice strip selects its track too.
- Steps 1–16: click to switch steps on/off. Above each step is a lane: drag it to set the step's velocity (VEL) or probability (PROB; steps below 100 % show their percentage).
- EUCLID: spreads a number of pulses evenly over 16 steps, with rotation.
- RND: random steps for this track (Shift: all tracks). CLR: clear the track. COPY: copy the pattern to the next slot.
- PLAY: runs LX-6's own clock when the host is stopped (standalone).
- Swing is a host-automatable parameter (Swing, 0–75 %). Mute/solo per voice are automatable too.

5. MIDI and automation

Note	Plays
C1 (36)	Drum 1
C#1 (37)	Drum 2
D1 (38)	Drum 3
D#1 (39)	Snare
E1 (40)	Cymbal
F1 (41)	Closed hi-hat

F#1 (42)	Open hi-hat

Every sound parameter, master, swing, tempo and mute/solo are host parameters, named like "Drum 1 Cutoff" or "Snare P Decay". Velocity from notes and steps is used by the voices.

6. Outputs

The LXR has two stereo outputs and LX-6 keeps them. Normally both are mixed to the main output. If you enable LX-6's Out 2 bus in your DAW, voices whose OUTPUT is St2/L2/R2 come out there instead, for separate processing. L/R settings send a voice to one side only.

7. LXR-02 hardware editor

If you own an Erica Synths LXR-02, LX-6 can edit and play it. Click LXR in the top bar:

- MODE: OFF; LXR ONLY (LX-6's own engine is silent, you hear the hardware); LX-6 + LXR (both play).
- MIDI OUTPUT: the port your LXR is connected to. With "None", messages only go to the plugin's MIDI output (route it in your DAW if it supports plugin MIDI out).
- PARAMETER CH: the channel for parameter changes. Set the LXR's global MIDI channel to the same number.
- Voice channels: the channel each track plays on (default 1–7). Set each voice's MIDI channel on the LXR to match.
- SEND KIT TO LXR: sends every parameter once, so the LXR sounds like LX-6.

Parameters 1–126 are sent as CC (CC number = parameter + 1), parameters 128 and up as NRPN (CC 99/98 + data entry CC 6). LFO and velocity destinations can only be set on the LXR itself. Hardware mode was built from the LXR firmware's MIDI code; it has not been tested on a physical LXR-02 yet.

8. Good to know

- One engine per project. The LXR engine is one piece of firmware, so only the first LX-6 in a project plays sound. More LX-6 instances say so in their top bar and still work as sequencer + LXR editor. Delete the first one and the next takes over.
- The engine runs at the LXR's own sample rate (44.14 kHz) and LX-6 converts to your project rate, so it sounds the same at 44.1, 48 or 96 kHz.
- Kits are small XML files (.lx6kit) you can share. Your sequencer patterns are saved with the DAW project.
- The window can be resized from the bottom-right corner.

9. Credits and licence

LX-6 by Ticco Ross — ticcoross.gumroad.com — ticcoross@gmail.com.

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